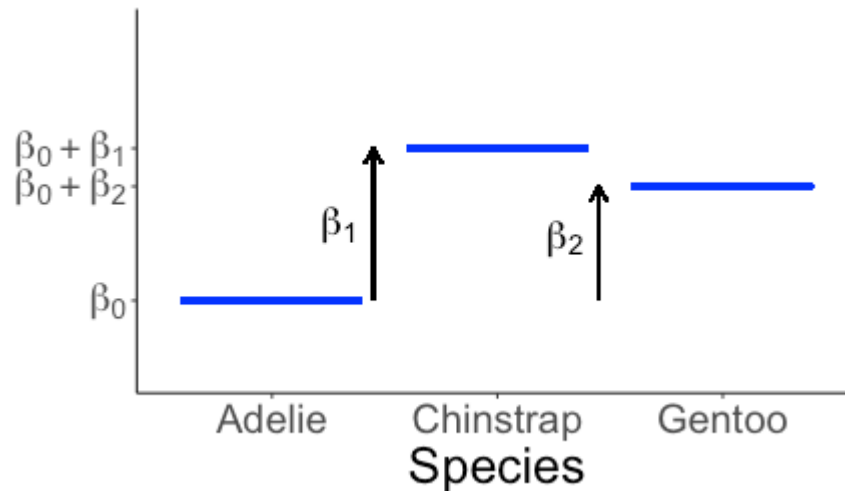


Regression with a categorical predictor

Reminders and agenda

- + Project part 2 due tonight
- + HW 4 due Monday
- + Today:
 - + Continue thinking about categorical explanatory variables
 - + How do we summarize strength of a relationship when there are more terms in the model?

Previously: regression with a categorical predictor

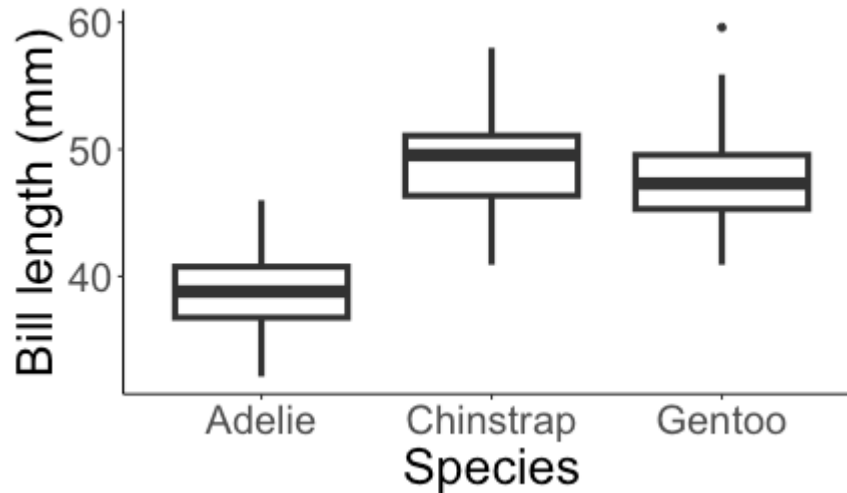


$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

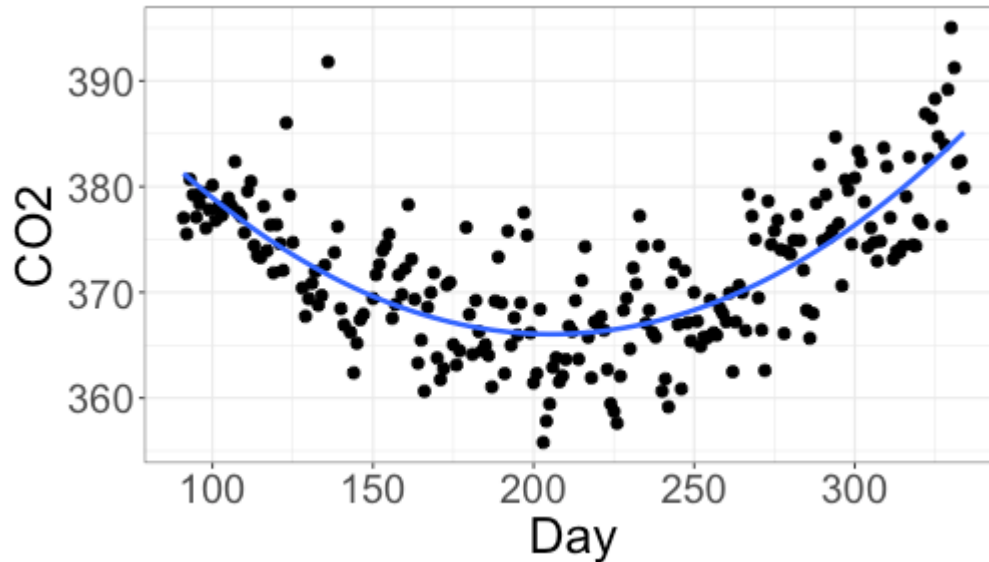
$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Previously: regression with a categorical predictor



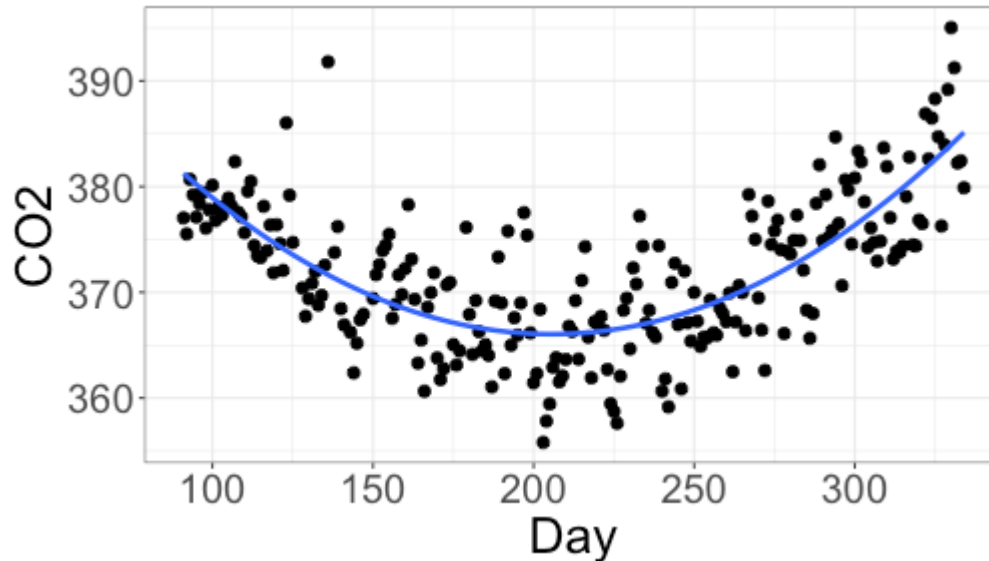
Can I use the correlation to summarize the relationship between species and bill length?

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Can I use the correlation to calculate the relationship between Day and CO2?

Measuring variability in the response

- + When we fit a regression model with least squares, we choose parameters to minimize the sum of squared residuals:

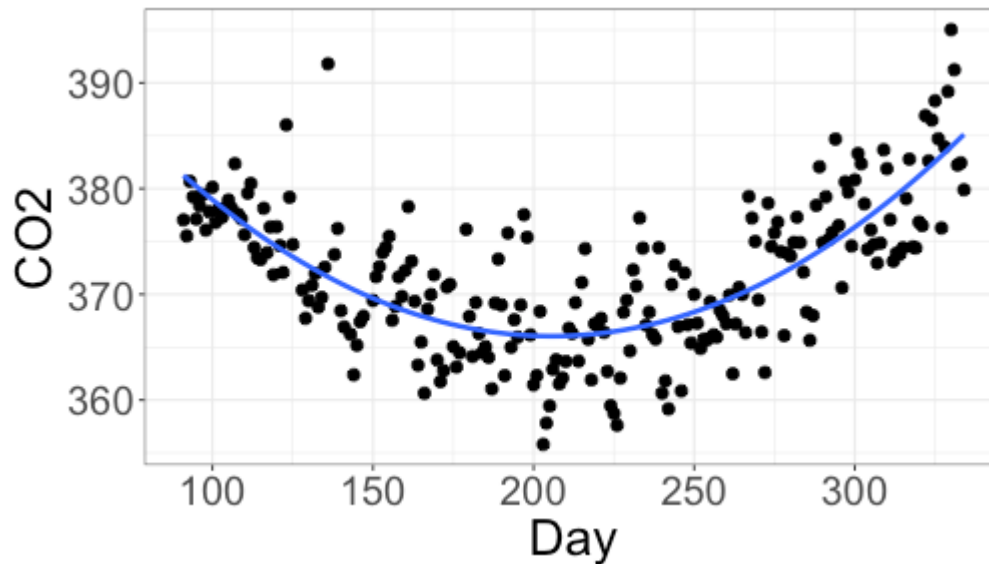
$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

- + We can think of SSE as the "amount of variability" in Y after predicting with our regression model
- + The amount of variability in Y *before* predicting with our regression model is

$$SSTotal = \sum_i (y_i - \bar{y})^2$$

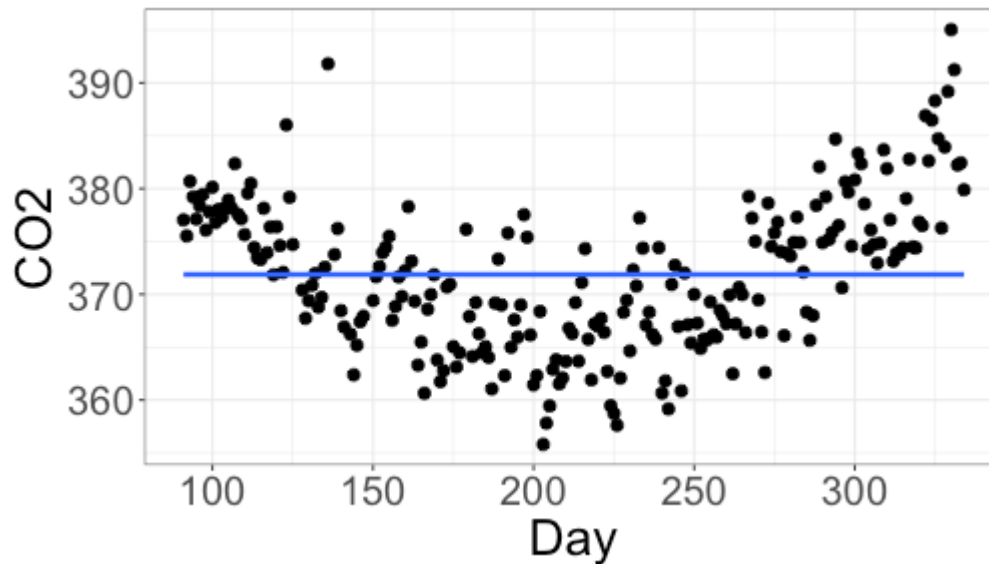
Thinking about SSE

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

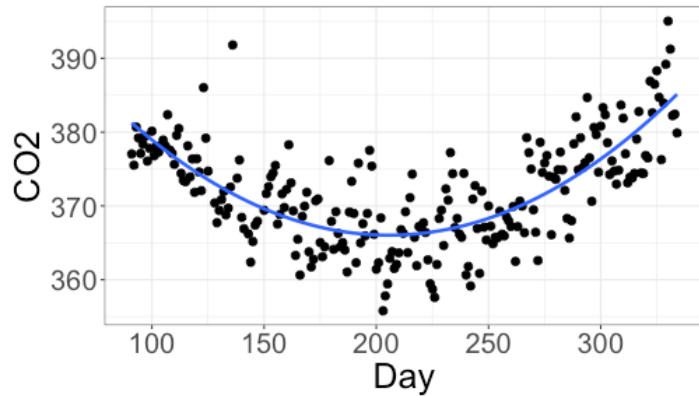


Thinking about $SSTotal$

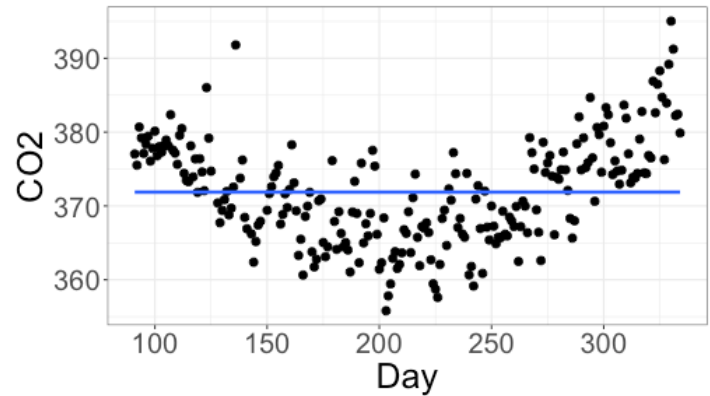
$$SSTotal = \sum_i (y_i - \bar{y})^2$$



SSE vs. SSTotal



SSE = 4992.71



SSTotal: 11703.69

Do I want SSE to be large relative to SSTotal, or small relative to SSTotal?

R^2 : the coefficient of determination

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

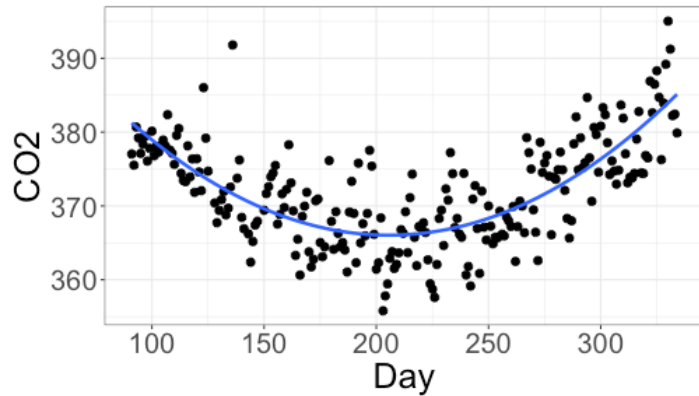
$$SSTotal = \sum_i (y_i - \bar{y})^2$$

The proportion of variability in Y "explained" by our regression model is

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

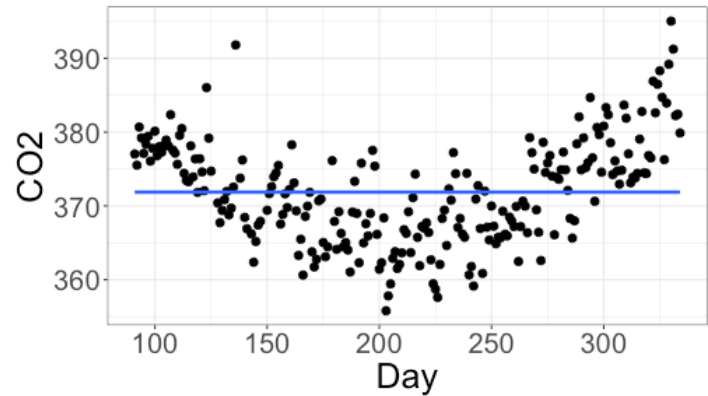
- + R^2 is between 0 and 1
- + A higher R^2 means our model is a "better fit" to the data
- + For regression with one variable, $R^2 = r^2$ (the squared correlation)

SSE vs. SSTotal



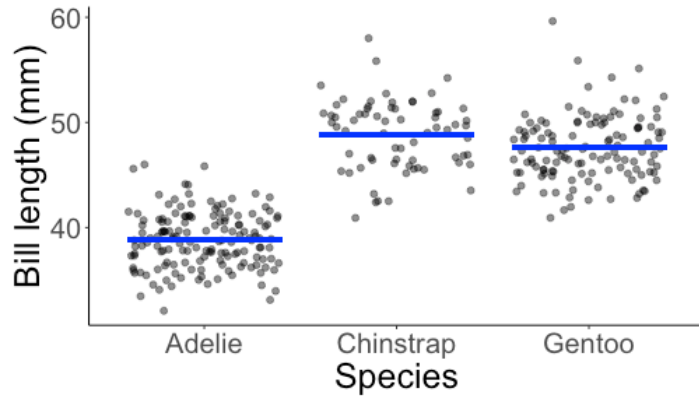
SSE = 4992.71

$$R^2 = 1 - \frac{4992.71}{11703.69} = 0.573$$



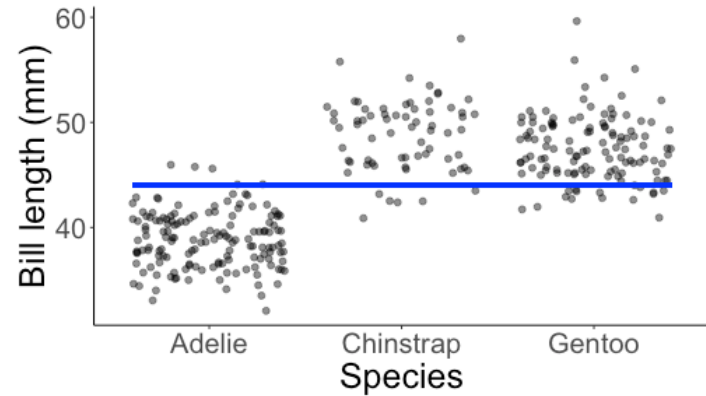
SSTotal: 11703.69

SSE vs. SSTotal



SSE = 2969.9

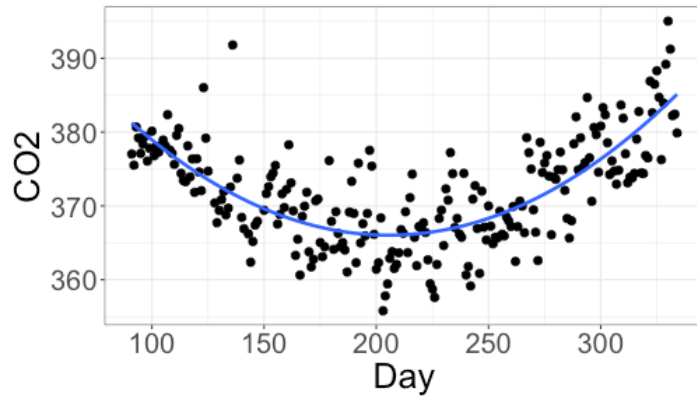
$$R^2 = 1 - \frac{2969.9}{10164.21} = 0.708$$



SSTotal = 10164.21

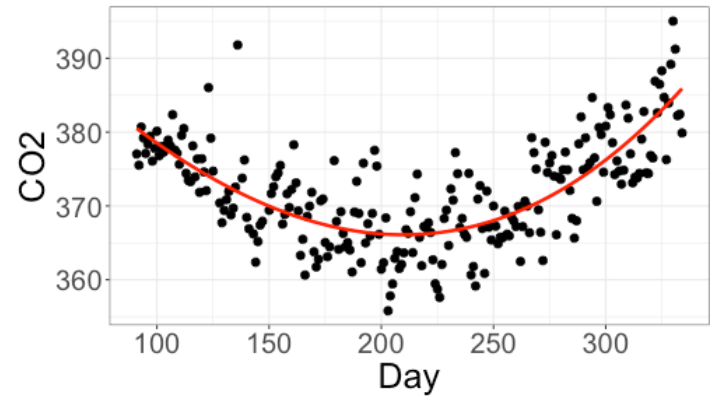
Comparing R^2 values

Polynomial of order 2:



$$R^2 = 0.573$$

Polynomial of order 3:



$$R^2 = 0.575$$

R^2 has increased very slightly -- does that mean that a polynomial of order 3 is a better fit?

R^2

R^2 will always increase when you add more parameters to the model.

Why? Because

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

and adding more parameters gives us more flexibility to decrease SSE.

Adjusted R^2

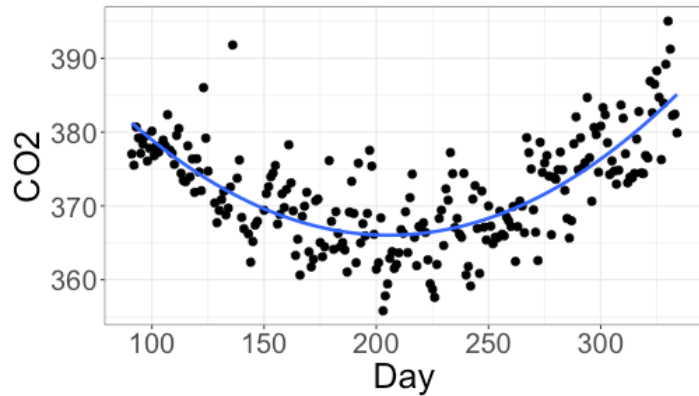
We can account for the number of parameters in the model with *adjusted R^2* :

$$R_{adj}^2 = 1 - \frac{SSE/(n - p)}{SSTotal/(n - 1)}$$

- + n = number of observations in data (sample size)
- + p = number of parameters (β terms) in model

Adjusted R^2

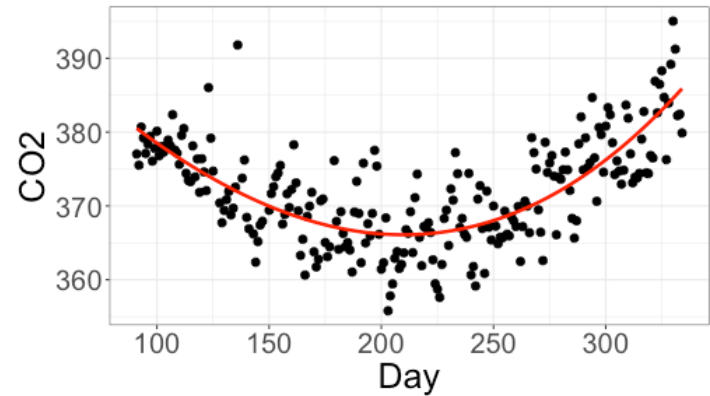
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



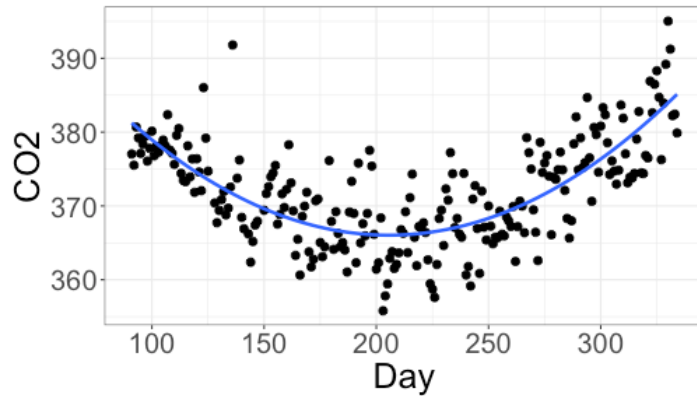
SSE = 4971.9

SSTotal = 11703.69

Why is SSTotal the same for the two models?

Adjusted R^2

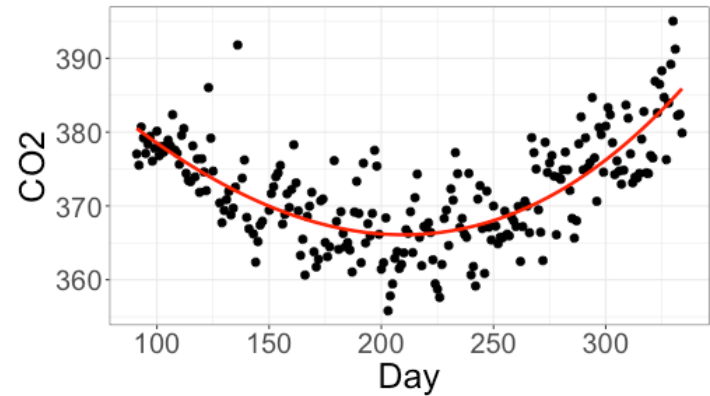
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



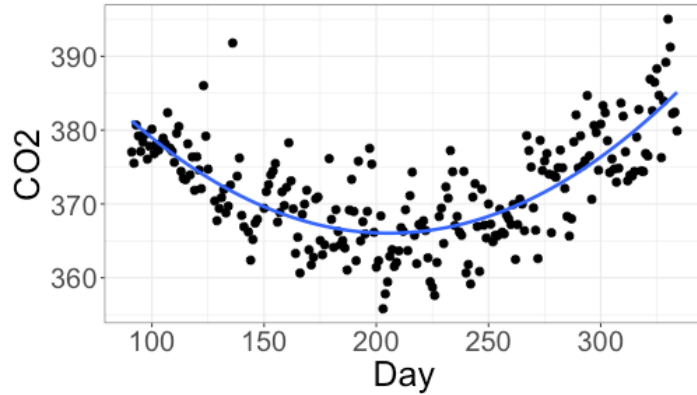
SSE = 4971.9

SSTotal = 11703.69

$$R_{adj}^2 = 1 - \frac{SSE/(n - p)}{SSTotal/(n - 1)}$$

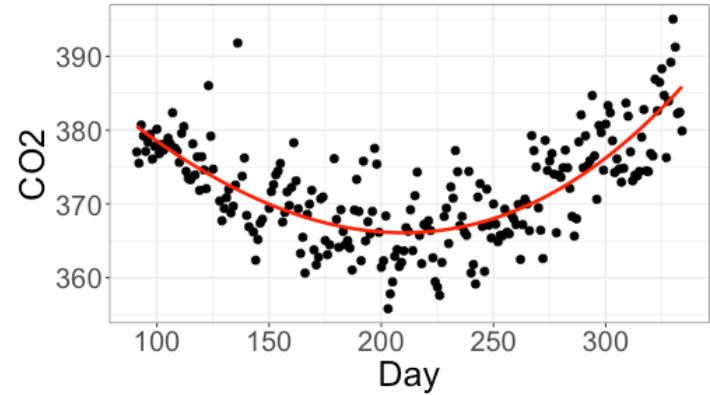
Adjusted R^2

Polynomial of order 2:



$$R^2_{adj} = 0.56976$$

Polynomial of order 3:



$$R^2 = 0.56972$$

Properties of R^2_{adj}

- + $R^2_{adj} < R^2$ (always)
- + R^2_{adj} accounts for the number of parameters in the model
- + R^2 can *never* decrease when predictors are added
- + R^2_{adj} *can* decrease if unhelpful predictors are added

Values in R

```
penguin_mod <- lm(bill_length_mm ~ species,  
                  data = penguins)  
summary(penguin_mod)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.7914	0.2409	161.05	<2e-16	***
speciesChinstrap	10.0424	0.4323	23.23	<2e-16	***
speciesGentoo	8.7135	0.3595	24.24	<2e-16	***
...					

Multiple R-squared: 0.7078, Adjusted R-squared: 0.7061

```
co2_mod <- lm(CO2 ~ poly(Day, 2, raw=TRUE),  
              data = CO2Germany)  
summary(co2_mod)
```

...

Multiple R-squared: 0.5734, Adjusted R-squared: 0.5698

Class activity

- + Practice regression with a single categorical predictor, with diamonds data
- + Calculate R^2 and R^2_{adj}
- + No need to submit this one on Canvas

https://sta112-f25.github.io/class_activities/ca_16.html